

Time and Change

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1 Time

- Time is ...
 - ▶ an important philosophical domain of inquiry;
 - ▶ a foundational ontology;
 - ▶ an essential domain in physics;
 - ▶ an essential domain in knowledge representation and reasoning;
 - ▶ a basic ingredient of most linguistic statements.
- The ontological nature of time must be carefully taken into account.

2 Source

- Elzbieta Hajnicz "Time Structures" (1996) (Sect.2 in particular)
- Additional references
 - ▶ John van Benthem (1980) *The Logic of Time*. Reidel, Dordrecht (Synthese Library 156). Revised and expanded edition 1991.
 - ▶ Allen, James F. *Maintaining knowledge about temporal intervals*. Communications of the ACM 26.11 (1983): 832-843.
 - ▶ Allen, James F., and Patrick J. Hayes. *Moments and points in an interval-based temporal logic*. Computational Intelligence 5.3 (1989): 225-238.
 - ▶ Davidson, Donald. "The logical form of action sentences (1967)." In: *Essays on actions and events*: 105-148. Oxford University Press, 2001.
 - ▶ Kamp, Hans. "Events, Instants and Temporal Reference". In: Bäuerle, R., Egli, U., von Stechow, A. (eds) *Semantics from Different Points of View*. Springer (1979)

3 Orders

- All the mereologies we considered are partial orders (reflexivity, transitivity, and antisymmetry).
- But orders are abstract structures that can model several (general and heterogeneous) comparative relations between entities:
 - ▶ “is more/less ... than”: “is bigger than”, “is smaller than”, “is later than” ...
 - ▶ “is to the left of”, “is an ancestor of”, “is a divisor of”, “is part of” ...
- A **partial order** is a binary relation $\leq$ such that:

$$\mathbf{a16} \quad x \leq x \qquad \qquad \qquad (\textit{reflexivity})$$

$$\mathbf{a17} \quad (x \leq y \wedge y \leq z) \rightarrow x \leq z \qquad \qquad \qquad (\textit{transitivity})$$

$$\mathbf{a18} \quad (x \leq y \wedge y \leq x) \rightarrow x = y \qquad \qquad \qquad (\textit{antisymmetry})$$

4 Additional properties for orders

d1 $x < y \triangleq x \leq y \wedge \neg(y \leq x)$ (*strict precedence order*)

a19 $x \leq y \vee y \leq x$ (*total / linear order*)

a20 $x < y \rightarrow \exists z(x < z \wedge z < y)$ (*dense order*)

a21 $x < y \rightarrow \exists z(x < z \leq y \wedge \neg \exists v(x < v < z))$ (*discrete order*)
 $x < y \rightarrow \exists z(x \leq z < y \wedge \neg \exists v(z < v < y))$

a22 $\exists x \forall y(x \leq y)$ (*left bounded*)

a23 $\exists x \forall y(y \leq x)$ (*right bounded*)

a24 $\forall x \exists y(y < x)$ (*left unbounded*)

a25 $\forall x \exists y(x < y)$ (*right unbounded*)

- On finite domains, all orders are discrete and bounded.
- **Examples:** $\langle \mathbb{N}, \leq \rangle$, $\langle \mathbb{Z}, \leq \rangle$, $\langle \mathbb{Q}, \leq \rangle$, $\langle \mathbb{R}, \leq \rangle$, $\langle 2^A, \subseteq \rangle$.

5 Instant theories of time

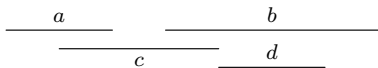
- Time is *absolute*, i.e. it is a container in which entities exist, are located, take place, or happen.
- **Domain:** punctual and extensionless temporal entities called *instants* or *moments*.
- **Primitive relation:** *precedence* ($<$ or $\leq$), i.e. a (strict) order relation between instants.

6 Structure of instants

- Total (linear) or partial (branching)
 - ▶ parallel times: alternative *worlds*
 - ▶ linear to the left, branching future: *planning*
 - ▶ linear to the right, branching past: *diagnostic*
- Bounded or unbounded
- Dense or discrete
 - ▶ commonsense?
 - ▶ Achilles and the turtle paradox
 - ▶ computers are discrete but calculus assume real time
- Continuity: the dividing instant. Not first-order definable!

7 Van Benthem theory of periods (1980)

- Time is *absolute*.
- **Domain**: extended temporal entities (*convex* stretches of time) called *intervals* or *periods*.
- **Primitive relations**: to take into account the extension of the periods Van Benthem introduces two primitive (order) relations:
 - ▶ *precedence* ($\prec$), and *parthood* ($\sqsubseteq$) also called *inclusion*.



- Similarly to the case of theories based on instants there is a huge space of possibilities.

8 Van Benthem theory of periods

- $\prec$ is a strict order that can be partially characterized by *reusing* some of the axioms already discussed, in particular $\prec$ is irreflexive and transitive (unboundness and discreteness can be added);
- $\sqsubseteq$ can be intuitively intended as a sort of parthood relation, in particular $\sqsubseteq$ is reflexive, transitive, and antisymmetric;

...but

- (i) there are some interactions between these primitives;
- (ii) the domain with convex intervals complicates the theory.

9 Van Benthem theory of periods

- MON:

$$\begin{aligned}\mathbf{a26} \quad & x \prec y \wedge z \sqsubseteq x \rightarrow z \prec y \\ & x \prec y \wedge z \sqsubseteq y \rightarrow x \prec z\end{aligned}$$

- To ensure the *convexity* of periods

$$\mathbf{a27} \quad x \prec y \wedge y \prec z \rightarrow \forall u((x \sqsubseteq u \wedge z \sqsubseteq u) \rightarrow y \sqsubseteq u)$$

- The *linearity* axiom becomes:

$$\mathbf{a28} \quad x \prec y \vee y \prec x \vee \exists z(z \sqsubseteq x \wedge z \sqsubseteq y)$$

10 Allen's theory of intervals (1983-89)

- Time is *absolute*, *unbounded* and *linear*.
- Domain:** convex intervals (aka periods)
- One primitive: *meets* ($\parallel$) that combines order and *adjacency*.
- 13 possible relations between any ordered pair of intervals (the six below + identity + the inverse relations):

Before $\frac{x}{\quad} \quad \frac{\quad}{y}$

Meets $\frac{x}{\quad} \frac{\quad}{y}$

Overlaps $\frac{x}{\frac{\quad}{y}}$

Starts $\frac{x}{\frac{\quad}{y}}$

During $\frac{\frac{x}{\quad}}{y}$

Finishes $\frac{\quad}{y} \frac{x}{\quad}$

11 Ontological priority

- From instants to intervals (periods)
 - ▶ $P = \{(x, y) \mid x, y \in I \text{ and } x < y\}$
 - ▶ $(x, y) \parallel (z, u)$ iff $y = z$
- From intervals (periods) to instants
 - ▶ $I = \{[x, y] \mid x, y \in P \text{ and } x \parallel y\}$ where
 $[x, y] = \{(z, v) \mid z, v \in P \text{ and } x \parallel z, v \parallel y\}$
 - ▶ $[x, y] < [z, u]$ iff $\exists v \in P$ such that $x \parallel v \parallel u$
- Can we establish a relation of "ontological priority" between instants and periods (Quine)?

12 Events as prior?

- *Relative approach*: time is an implicit structure induced by temporal relations over *events*.
 - ▶ (Leibniz vs. Newton)
- Simultaneity is not identity.
- Commonsense, psychological and linguistic evidence.

13 Kamp's theory of events (1979)

- **Domain:** *events*.
- **Primitives:** *precedence* ($\triangleleft$) and (temporal) *overlap* (O) between events.
- The precedence is a strict partial order and the overlap is reflexive and symmetric relation.
- In addition one needs axioms linking the two primitives:

$$\mathbf{a29} \quad x \triangleleft y \rightarrow \neg O(x, y)$$

$$\mathbf{a30} \quad x \triangleleft y \wedge O(y, z) \wedge z \triangleleft t \rightarrow x \triangleleft t$$

$$\mathbf{a31} \quad x \triangleleft y \vee O(x, y) \vee y \triangleleft x$$

14 The need for events

- Linguistic evidence: event names, event anaphora, verb modification...

15 The need for events in semantics: Davidson (1967)

- How many arguments for a verb? (verb modification)
 - ▶ “Jones buttered the toast”
 $Butter(Jones, toast)$
 - ▶ “Jones buttered the toast with the knife”
 $Butter(Jones, toast, knife)$
 - ▶ Many different predicates of different arity; reasoning requires many axioms
- Events as entities in the domain, i.e., event reification
 - ▶ $\exists e Butter(e, Jones, toast)$
 - ▶ $\exists e (Butter(e, Jones, toast) \wedge With(e, knife))$
 - ▶ A single predicate, entailment for free

16 Generalisation: neo-davidsonian approach

- T. Parsons (1990)
- No distinction between “core” participants and the rest
- $\exists e (Butter(e) \wedge Agent(e, Jones) \wedge Patient(e, toast) \wedge Instrument(e, knife))$
- Predicates for thematic roles

17 The need for events

- Linguistic evidence: event names, event anaphora, verb modification...
- Causal reasoning and planning are based on events, esp. actions
- But: identity criteria for events are not obvious
 - ▶ spatio-temporal co-localization (again, simultaneity is not identity)
 - myself vs. my life
(distinction between objects and events)
 - the spinning of the ball vs. the warming up of the ball
(events are not identified by the objects participating in it)
 - ▶ causal equivalence, logical equivalence...

18 Change: the method of temporal arguments

- New kind of entities called *times*
- New primitive $\text{Ex}(x, t)$: “the object x *exists* at time t ”
 - ▶ different from $\exists$.
 - ▶ not needed in domains without births, creations, deaths, ...
- Temporally qualify all the (possibly) *time variant* primitives applying to objects existing in time:
 - ▶ e.g. $\text{White}(x, t)$ (“at time t , the object x is white”) or $\text{Supp}(x, y, t)$ (“at time t , object x supports object y ”)
- Change:
 - ▶ e.g. $\text{White}(x, t_1) \wedge \text{Black}(x, t_2)$

19 Endurantism vs. Perdurantism

- Objects undergoing the change can *endure*, i.e. they can be *wholly present* at any time at which they exist, they maintain their identity through change.
 - ▶ Endurantists may also accept perduring objects, e.g., *events*.
- All the objects persist by *perduring*, i.e. objects are extended in time (as they do in space) by having different (*temporal*) *parts/slices* (max. part of the object existing at t) at different times.

20 Perdurantism and change of intrinsic properties

- To capture the fact that two objects have incompatible intrinsic properties at different times, endurantists need to *temporally qualify* such properties (as seen) or the exemplification relation while perdurantists can rely on temporal parts.
- Temporal qualification is not necessary for perdurantists given that objects have different temporal parts at different times ($x@t$ is the temporal part of the object x at time t ;)
 - ▶ $\text{White}(x@t_1) \wedge \text{Black}(x@t_2)$ instead of $\text{White}(x, t_1) \wedge \text{Black}(x, t_2)$

21 Perdurantism vs. Endurantism

- The distinction between perdurantism and endurantism is informally stated in terms of the notions of being *partially* or *wholly* present.
- Being partially present has been quite precisely characterized while being wholly present is still quite obscure (alternative attempts exist).
- Perdurantism vs. endurantism often reduces to positions on *parthood*:
 - ▶ endurantists, rejecting the existence of temporal slices at least for some kinds of individuals, claim that a primitive temporally qualified parthood (*temporary parthood*) is required
 - ▶ perdurantists assume an atemporal parthood (*parthood simpliciter*, or simply *parthood*) is enough.

22 Parthood simpliciter vs. temporary parthood

Parthood simpliciter

$\text{Ex}(x, t)$ “ x exists at t ”;

$\text{P}(x, y)$ “ x is part of y ”.

Temporary Parthood

$\text{Ex}(x, t)$ “ x exists at t ”;

$\text{tP}(x, y, t)$ “ x is part of y at t ”.

Definitions on the basis of P:

d2 $\text{O}(x, y) \triangleq \exists z(\text{P}(z, x) \wedge \text{P}(z, y))$

d3 $\text{TP}(x, y, t) \triangleq \text{Ex}(x, t) \wedge \text{Ex}(y, t) \wedge \neg \exists t'(\text{Ex}(x, t') \wedge t' \neq t) \wedge \text{P}(x, y) \wedge \forall z(\text{P}(z, y) \wedge \text{Ex}(z, t) \rightarrow \text{O}(z, x))$

d4 $\text{tP}(x, y, t) \triangleq \exists zw(\text{TP}(z, x, t) \wedge \text{TP}(w, y, t) \wedge \text{P}(z, w))$

Definitions on the basis of tP:

d5 $\text{tO}(x, y, t) \triangleq \exists z(\text{tP}(z, x, t) \wedge \text{tP}(z, y, t))$

d6 $\text{tTP}(x, y, t) \triangleq \neg \exists t'(\text{Ex}(x, t') \wedge t' \neq t) \wedge \text{tP}(x, y, t) \wedge \forall z(\text{tP}(z, y, t) \rightarrow \text{tO}(z, x, t))$

d7 $\text{P}(x, y) \triangleq \forall t(\text{Ex}(x, t) \rightarrow \text{tP}(x, y, t))$

23 The theories $\mathcal{T}_{\text{tP}}$ and $\mathcal{T}_{\text{P}}$

$\mathcal{T}_{\text{tP}}$: **temporary parthood** (Sider)

$\mathcal{T}_{\text{P}}$: **parthood simpliciter**

$$\mathbf{a32} \quad \exists t(\text{Ex}(x, t))$$

$$\mathbf{a33} \quad \text{tP}(x, y, t) \rightarrow \text{Ex}(x, t) \wedge \text{Ex}(y, t)$$

$$\mathbf{a34} \quad \text{Ex}(x, t) \rightarrow \text{tP}(x, x, t)$$

$$\mathbf{a35} \quad \text{tP}(x, y, t) \wedge \text{tP}(y, z, t) \rightarrow \text{tP}(x, z, t)$$

$$\mathbf{a36} \quad \text{Ex}(x, t) \wedge \text{Ex}(y, t) \wedge \neg \text{tP}(x, y, t) \rightarrow \\ \exists z(\text{tP}(z, x, t) \wedge \neg \text{tO}(z, y, t))$$

$$\mathbf{pd} \quad \text{Ex}(x, t) \rightarrow \exists y(\text{tP}(y, x, t))$$

$$\mathbf{a32} \quad \exists t(\text{Ex}(x, t))$$

$$\mathbf{a37} \quad \text{P}(x, x)$$

$$\mathbf{a38} \quad \text{P}(x, y) \wedge \text{P}(y, x) \rightarrow x = y$$

$$\mathbf{a39} \quad \text{P}(x, y) \wedge \text{P}(y, z) \rightarrow \text{P}(x, z)$$

$$\mathbf{a40} \quad \neg \text{P}(x, y) \rightarrow \exists z(\text{P}(z, x) \wedge \neg \text{O}(z, y))$$

$$\mathbf{a41} \quad \text{P}(x, y) \wedge \text{Ex}(x, t) \rightarrow \text{Ex}(y, t)$$

$$\mathbf{pdn} \quad \text{Ex}(x, t) \rightarrow \exists y(\text{P}(y, x, t))$$

- Given (d4), in $\mathcal{T}_{\text{P}}$, all the axioms of $\mathcal{T}_{\text{tP}}$ can be recovered.
- Given (d7), in $\mathcal{T}_{\text{tP}}$, (a38) cannot be recovered (P is not antisymmetric); one needs:

$$\mathbf{a42} \quad \forall t(\text{Ex}(x, t) \rightarrow \text{tP}(x, y, t)) \wedge \forall t(\text{Ex}(y, t) \rightarrow \text{tP}(y, x, t)) \rightarrow x = y$$

- The unicity of the temporal part is not guaranteed
(model with a, b existing at a single time t such that $(a, b, t), (b, a, t) \in \text{tP}^I$).
[More generally, problem of the definition of temporal part]

ADDITIONAL SLIDES

24 Allen's theory of periods (1983-89)

$$\mathbf{a43} \quad x\|y \wedge x\|z \wedge u\|y \rightarrow u\|z$$

$$\mathbf{a44} \quad x\|y \wedge z\|u \rightarrow (x\|u \bar{\vee} \exists v(x\|v\|u) \bar{\vee} \exists v(z\|v\|y))$$

$$\mathbf{a45} \quad \forall x \exists y z (y\|x \wedge x\|z)$$

$$\mathbf{a46} \quad x\|y \rightarrow \exists z uv (z\|x\|y\|u \wedge z\|v\|u)$$

$$\mathbf{a47} \quad x\|y\|z \wedge x\|u\|z \rightarrow y = u$$

$$\mathbf{t1} \quad \neg x\|x$$

$$\mathbf{t2} \quad x\|y \rightarrow \neg \exists z (x\|z\|y)$$

$$\mathbf{t3} \quad x\|y \rightarrow \neg y\|x$$

$$\mathbf{t4} \quad x\|y \wedge z\|y \wedge x\|u \rightarrow z\|u$$

$$\mathbf{t5} \quad x\|y \wedge x\|z \wedge u\|x \rightarrow u\|z$$

25 From instants to intervals/periods (1)

- Consider $\langle I, < \rangle$ where $<$ is irreflexive, transitive, linear, left/right unbound.
- $P = \{(x, y) \mid x, y \in I \text{ and } x < y\}$
- $(x, y) \parallel (z, u)$ iff $y = z$

t6 $(x_1, x_2) \parallel (y_1, y_2) \wedge (x_1, x_2) \parallel (z_1, z_2) \wedge (u_1, u_2) \parallel (y_1, y_2) \rightarrow (u_1, u_2) \parallel (z_1, z_2)$

Proof. We need to prove that $u_2 = z_1$. From the hypotheses and the def. of $\parallel$, $x_2 = y_1$, $x_2 = z_1$, $u_2 = y_1$. The thesis trivially follows. $\square$

t7 $(x_1, x_2) \parallel (y_1, y_2) \wedge (z_1, z_2) \parallel (u_1, u_2) \rightarrow ((x_1, x_2) \parallel (u_1, u_2) \vee \exists (v_1, v_2)((x_1, x_2) \parallel (v_1, v_2) \parallel (u_1, u_2)) \vee \exists (v_1, v_2)((z_1, z_2) \parallel (v_1, v_2) \parallel (y_1, y_2)))$.

Proof. Consider x_2 and u_1 . By linearity, $x_2 = u_1 \vee x_2 < u_1 \vee u_1 < x_2$ (it easy to see that these are xor). If $x_2 = u_1$ then $(x_1, x_2) \parallel (u_1, u_2)$; if $x_2 < u_1$, then (x_2, u_1) exists and $(x_1, x_2) \parallel (x_2, u_1) \parallel (u_1, u_2)$; if $u_1 < x_2$, then (u_1, x_2) exists and given that, by the hypotheses, $z_2 = u_1$ and $x_2 = y_1$, $(z_1, z_2) \parallel (u_1, x_2) \parallel (y_1, y_2)$. $\square$

26 From instants to intervals/periods (2)

- $P = \{(x, y) \mid x, y \in I \text{ and } x < y\}$

- $(x, y) \parallel (z, u) \text{ iff } y = z$

t8 $\exists(y_1, y_2)(z_1, z_2)((y_1, y_2) \parallel (x_1, x_2) \wedge (x_1, x_2) \parallel (z_1, z_2))$

Proof. Consider x_1 , by left unboundedness, $\exists a(a < x_1)$, then $(a, x_1) \parallel (x_1, x_2)$. Similarly for the right unboundedness. $\square$

t9 $(x_1, x_2) \parallel (y_1, y_2) \rightarrow \exists(z_1, z_2)(u_1, u_2)(v_1, v_2)((z_1, z_2) \parallel (x_1, x_2) \parallel (y_1, y_2) \parallel (u_1, u_2) \wedge (z_1, z_2) \parallel (v_1, v_2) \parallel (u_1, u_2))$

Proof. From the hypotheses, $x_1 < x_2 = y_1 < y_2$ that, by transitivity, implies $x_1 < y_2$. Consider x_1, y_2 , by left/right unboundedness, $\exists ab(a < x_1 \wedge y_2 < b)$. It is enough to consider $(z_1, z_2) = (a, x_1)$, $(u_1, u_2) = (y_2, b)$, and $(v_1, v_2) = (x_1, y_2)$. $\square$

t10 $(x_1, x_2) \parallel (y_1, y_2) \parallel (z_1, z_2) \wedge (x_1, x_2) \parallel (u_1, u_2) \parallel (z_1, z_2) \rightarrow (y_1, y_2) = (u_1, u_2)$

Proof. From the def. of $\parallel$, $x_2 = y_1 = u_1$ and $z_1 = y_2 = u_2$. The thesis follows directly from the definition of P . $\square$

27 From intervals/periods to instants

- Consider a structure $\langle P, \parallel \rangle$ where $\parallel$ satisfies (a43)-(a47).
- $I = \{[x, y] \mid x, y \in P \text{ and } x \parallel y\}$ where $[x, y] = \{(z, v) \mid z, v \in P \text{ and } x \parallel z, v \parallel y\}$
- $[x, y] < [z, u]$ iff $\exists v \in P$ such that $x \parallel v \parallel u$

t11 $\neg[x, y] < [x, y]$

Proof. From the def. of $[z, v]$ and $<$, $\exists z(x \parallel z \parallel y) \wedge x \parallel y$ against (t2). $\square$

t12 $[x, y] < [z, v] \wedge [z, v] < [u, w] \rightarrow [x, y] < [u, w]$

Proof. From the def. of $[z, v]$ and $<$, $\exists k j(x \parallel k \parallel v \wedge z \parallel j \parallel w)$ and $z \parallel v$. From $z \parallel v$, $k \parallel v$, $z \parallel j$, by (a43), $k \parallel j$. From $k \parallel j$, by (a46), $\exists abc(a \parallel k \parallel j \parallel b \wedge a \parallel c \parallel b)$ and then, given that $x \parallel k$ and $j \parallel w$, by (t4) and (t5), $\exists c(x \parallel c \parallel w)$. $\square$

t13 $\forall [x, y] \exists [z, v] ([z, v] < [x, y]) \quad [\sim \text{ for } [x, y] < [z, v]]$

Proof. Consider x , by (a45), $\exists k(k \parallel x)$ and then we can consider $[k, x]$. We directly have $[k, x] < [x, y]$ because $k \parallel x \parallel y$. $\square$

t14 $[x, y] < [z, v] \vee [z, v] < [x, y] \vee [x, y] = [z, v]$

Proof. From df. $[-, -]$, $x \parallel y \wedge z \parallel v$, by (a44), $(x \parallel v \bar{\vee} \exists k(x \parallel k \parallel v) \bar{\vee} \exists k(z \parallel k \parallel y))$. From $x \parallel v$ we have $[x, y] = [z, v]$; from $\exists k(x \parallel k \parallel v)$ we have $[x, y] < [z, v]$; from $k(z \parallel k \parallel y)$ we have $[z, v] < [x, y]$. $\square$